

CA3 – Turing machine

1. As you know, the language $L1 = \{ww \mid w \text{ is in } \{0, 1\}^*\}$ is not a context-free language (CFL). Considering $L1$, answer to the following questions:
2. Show a TM able to recognize $L1$;
(4 Pontos)
3. Show the trace of computation of your TM when the input is 0101; 
(0.5 Pontos)
4. Show the trace of computation of your TM when the input is 10; 
(0.5 Pontos)
5. TOPIC B. It is known that a PDA with two stacks is, in terms of computing power, equivalent to a Turing Machine. Consider the language $L2 = \{w\#w \mid w \text{ is in } \{0, 1\}^*\}$ and answer to the following questions: 
6. Show a possible PDA with two stacks that can implement $L2$. Describe the transition function that you propose for your extended PDA;
(4 Pontos)
7. Show the processing steps with the instantaneous descriptions of your PDA when the input is 01#01;
(1 Ponto)